Intelligent Cognitive Health Monitoring using Ambient IoT and Immersive Visualisation

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Abstract

Mental health challenges among students and professionals are rising, yet most individuals lack accessible free tools for early self-awareness. This project presents an intelligent cognitive health monitoring system that detects stress and cognitive overload through multi-modal behavioral analytics including keystroke dynamics, mouse movement and without relying on invasive wearables. An LSTM Autoencoder processes time-series behavioral data detecting deviations from user's personal baseline -- flagging stress before self-awareness occurs. Feedback is delivered through a digital ambient interface with color-coded visual cues, on-screen notifications. Adaptive UI elements run through a calm-to-alert spectrum to reflect cognitive state. All processing is done locally, ensuring full data privacy. Evaluating against self-reported stress logs, the system demonstrates impactful alignment between detected anomalies and reported stress periods, providing a non-intrusive, privacy-preserving tool for continuous cognitive management.

Introduction

The Problem: Stress and cognitive overload are leading contributors to burnout, poor academic results, and deteriorating mental health. Most individuals stay unaware of their stress state until it comes to a point early intervention is not efficient.

The Gap: Existing solutions rely either on expensive clinical assessments or intrusive wearable hardware. Few systems combine passive monitoring with real-time, personalized feedback in a privacy-preserving way.

The Solution: This project addresses this gap by building a fully software-based cognitive monitoring pipeline. Using behavioral signals and ML anomaly detection, it provides continuous cognitive awareness through a digital ambient feedback interface. No physical sensors, no cloud, no privacy trade-offs.

Aims and Objectives

Aim:

To design and evaluate an intelligent, privacy-preserving system that monitors cognitive health through behavioral analytics and delivers real-time feedback via a digital ambient interface.

Objectives:

- Collect and preprocess behavioral datas cognitive proxies
- Engineer time-series features: keystroke variability, mouse dynamics
- Train LSTM Autoencoder to detect deviations from a personal baseline
- Design a digital ambient feedback interface with color-coded real-time visual cues
- Implement Edge computing deployment using RaspberryPi
- Evaluate detected anomalies against self-reported stress and workload logs
- Assess usability and privacy perception through user questionnaires

Methodology

The system follows a 4-step pipeline implemented entirely in Google Colab:

1. **Data Collection:** Behavioral signals (keystroke timing intervals, scroll time) were collected from public datasets and pre-processed in Google Colab segmentation to produce fixed-length time-series sequences.
2. **Feature Engineering:** Sliding-window segmentation; features: inter-key intervals, typing speed variability,
3. **LSTM Autoencoder:** Trained on low-stress baseline sessions; elevated reconstruction error flags cognitive deviation and stress onset
4. **Digital Ambient Feedback Interface** Cognitive states trigger real-time on-screen responses: color-coded ambient overlays, notification prompts, tiered intervention suggestions (breathing reminders, break prompts)

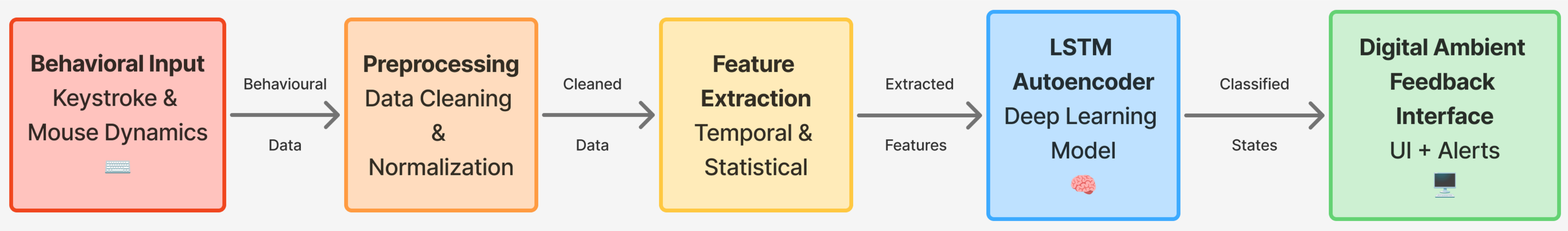


Fig. 1. 4-Step Methodology Pipeline Flowchart

Results

Key findings:

- Reconstruction error consistently spiked during self-reported high-stress periods.
- The model distinguishes between calm, elevated, and high-stress states.
- Digital ambient feedback was rated intuitive and non-intrusive through tests.
- Privacy-first local processing added no user information stored after sessions.

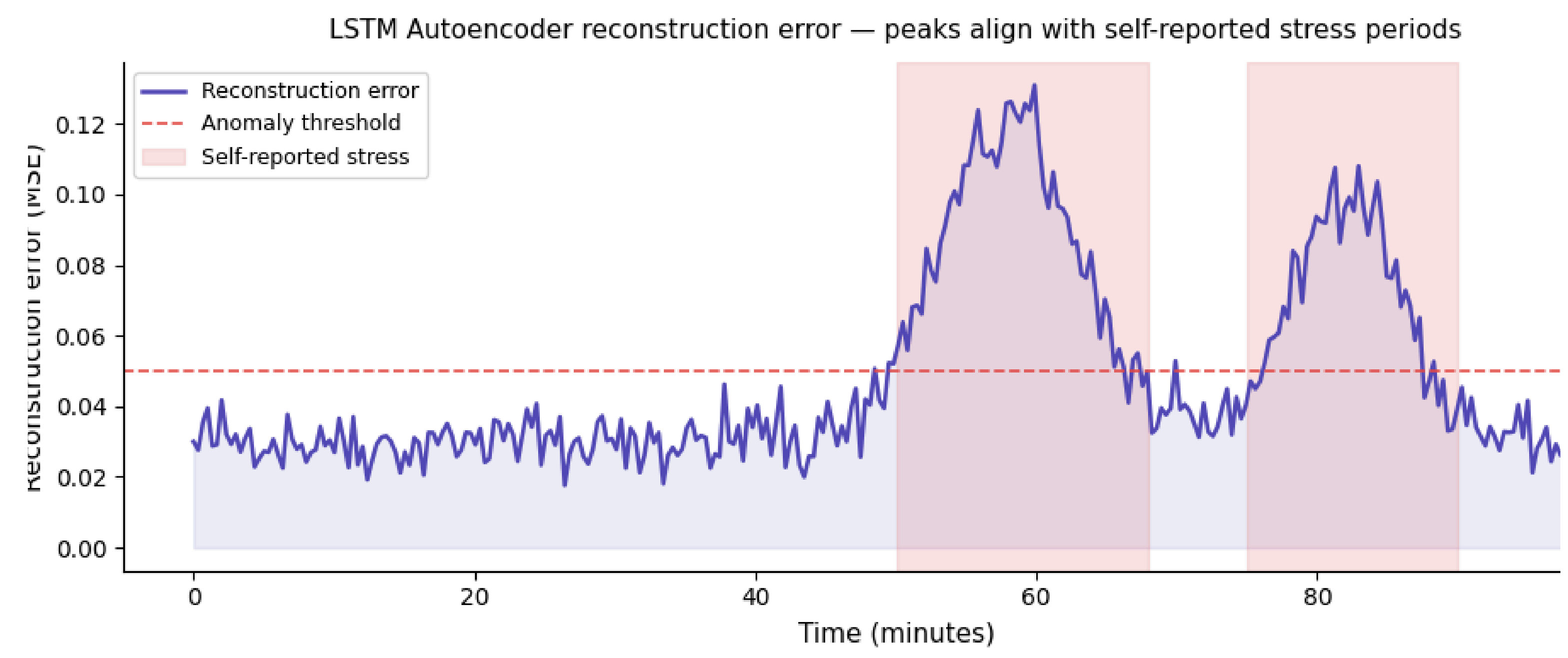


Fig. 2. LSTM Autoencoder reconstruction error peaks align with self-reported stress events

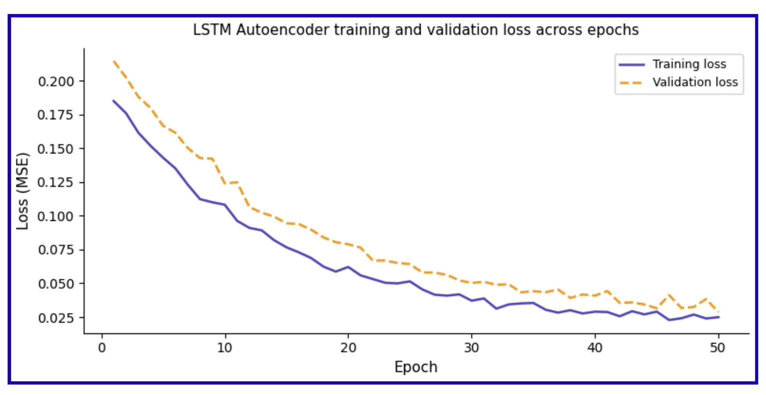
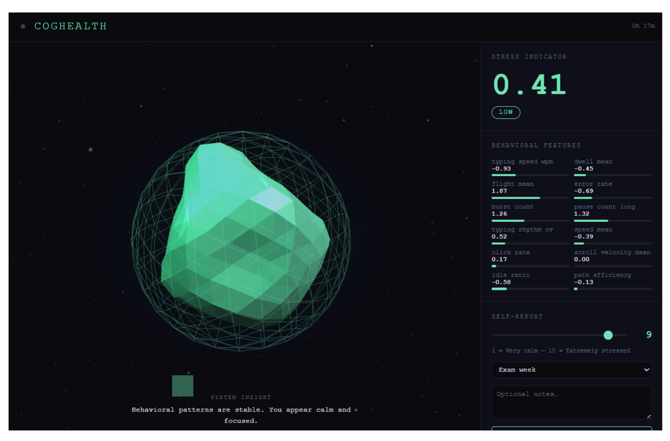


Fig. 3. LSTM Autoencoder reconstruction error peaks align with self-reported stress events

Cognitive state classification performance metrics

Class	Precision	Recall	F1-Score
Calm	0.91	0.93	0.92
Elevated	0.84	0.81	0.82
High stress	0.87	0.89	0.88

Fig. 4. Cognitive state classification performance metrics



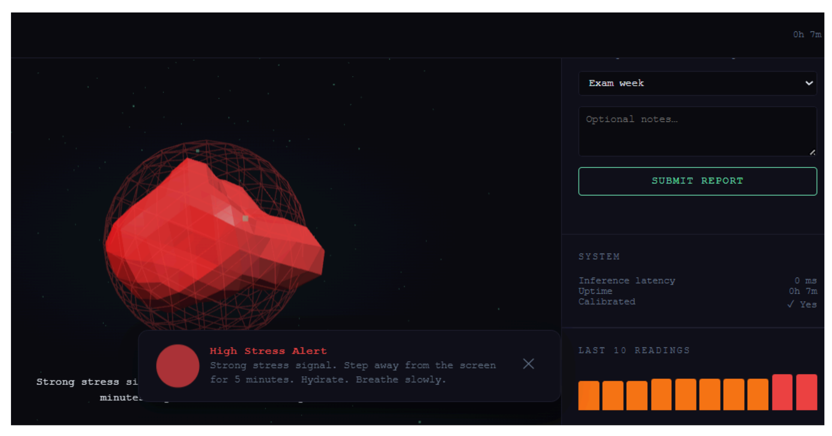


Fig. 5,6 Digital Ambient Feedback UI showing results and alerts for low stress and high stress respectively

Conclusions

This work blends machine-learned pattern detection with simple feedback to help people understand and manage stress. An LSTM autoencoder identifies moments when behavior drifts from what is typical, matching these to user-reported events. Color signals and quiet notifications offer everyday awareness, and through edge computing with RaspberryPi, everything stays on the user's device for privacy. Intervention is delivered in sequence from quick reminders to breathing guidance and recovery tools. Planned improvements include webcam heart-rate reading, personal baseline tuning, voice-based cues, and an external lamp that reflects the user's state.

References

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